

Introduction notes on Signals

信号の紹介

1. Be able to explain the **characteristics** of **analog to digital signals**.
2. Explaining conversion method (**PCM**)

a. 本科目「信号処理 "signal processing" の概要を把握し

Understand the outline of this subject, "Signal Processing," and

b. アナログ信号 "analog signal" と デジタル信号 "digital signal" について

understand the characteristics of analog signals and digital signals,

c. 両者間の変換方法を理解しよう

as well as methods of converting between the two.

本科目の概要

Overview / Outline of the subject

アナログ信号

Analog Signal

周波数スペクトル

Frequency Spectrum

ディジタル信号

Digital Signal

A/D変換, D/A変換

A/D & D/A Conversion

PCM の基礎

Basics of PCM

取扱対象：デジタル音声信号

1. Dealing with : **digital audio signals.**

取扱事項：音声データの入出力・生成・加工・分析

2. Matters handled : audio data input / output , generation , processing and analysis of data

一般的なデジタル音声メディアの仕組

3. General structure of digital audio media / How general digital audio media works

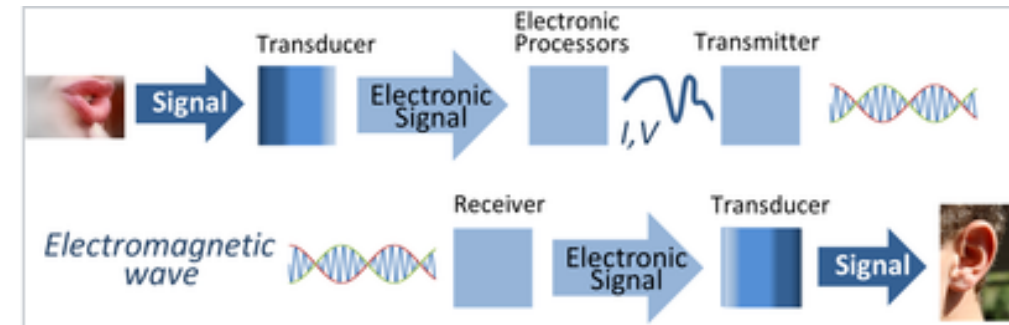
- a. 音波 "acoustic wave, sound wave" (原信号 "original signal" / 再生信号 "reproduced signal")

【音響電気変換(マイクログフォン／スピーカ)】

- b. Acoustoelectric conversion (microphone / speaker)

- c. アナログ信号波 "analog signal wave"

電圧・電流の波. 物理的な現象. Waves of voltage and current. Physical phenomenon.



d. **A/D変換／D/A変換**

A/D Conversion / D/A Conversion

e. **デジタルデータ** "digital data" (バイナリデータ, ビット列)

binary data , bit sequence

物理的な現象でもあるが, 論理的な存在として扱う.

Although it is physical phenomenon , but treated as logical entity.

f. **加工, 通信, 記録／再生**

Processing , communication , recording / playback

g. **デジタル信号波** "digital signal wave", パルス "pulse train"

デジタル信号波と云いつつ実体はアナログ信号波でもある. 電圧・電流とかレーザとか物理的な現象なので.

Although called digital signal waves , they are in fact also analogue signal waves. It is a physical phenomenon such as voltage , current laser etc.

アナログ信号処理: ハードウェア(電気・電子回路)による信号波に対する処理

Analog signal processing : processing of signal waves by hardware

応用例: TV(昭和期のNTSC), ラジオ(AM, FM), テープレコーダ(カセット, VHS), 電子楽器(エレキギター, エフェクタ, シンセサイザ, 等), ...

Eg. TV (NTSC), radio (AM, FM), tape recorder (cassette, VHS) electronic musical instruments (electric guitar, effector, synthesizer)

アナログ信号の特徴: 生成・変換の直後には原信号に忠実であっても, 伝送・記録に伴い劣化し易い.

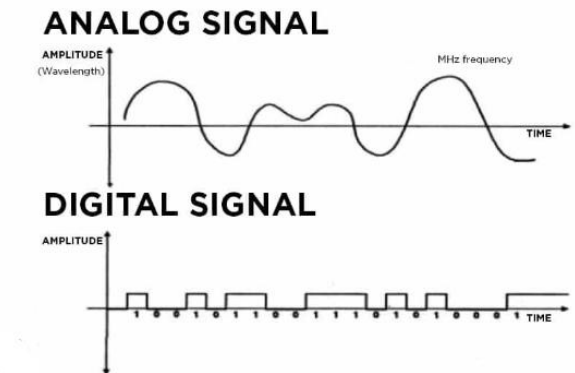
Characteristics of analog signals: Even if they are faithful to the original signal immediately after generation and conversion, they are easily degraded during transmission and recording.

a. 時間的にも強度的にも連続的 "continuous" → 情報量は無限 "infinite"

→ In time and intensity "continuous" → Infinite amount of information

b. 劣化の種類: 波形の歪 "waveform distortion", 雑音 "noise" の混入

Types of distortion → waveform distortion and inclusion of noise in the waveform



c. 記録による劣化：記録媒体の物理的な変化 —deterioration due to recording ; physical changes in the recording medium

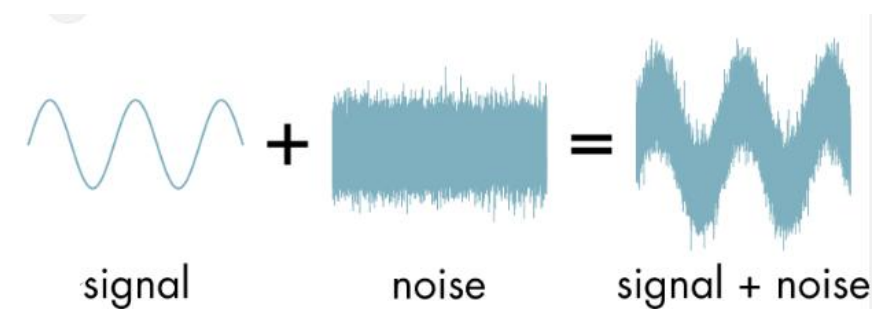
具体例：磁気テープの伸縮→歪, レコード盤の傷→雑音, 等

Examples ; Expansion and Contraction of magnetic tape , distortion , scratches on vinyl records, noise etc. .

d. 伝送による劣化：伝送路の周波数特性 - distortion due to transmission ; Frequency response of transmission lines

具体例：周波数による伝わりやすさ(速度や減衰)の差異→ 伝送の前後でスペクトルが変化→歪, 混信→ 雑音

Example ; Difference in ease of transmission (speed and attenuation) due to frequency → Spectrum changes before and after transmission → distortion , interference → Noise



a. 周波数スペクトル "frequency spectrum": 信号 $X(t)$ を構成する周波数成分の分布 $X(w)$

Frequency Spectrum: Distribution of the frequency components that make up a signal

b. フーリエ級数展開 "Fourier series expansion": 任意の周期波形 = 1/k 周期の正弦波の和

Fourier series expansion : Sum of sinusoids of arbitrary periodic waveforms = 1/k period

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \exp(j\omega_k t), \quad \omega_k = \frac{2\pi k}{T}$$

c. フーリエ変換 "Fourier transform": 任意の波形 = あらゆる周波数の正弦波の和

Fourier Transform : Arbitrary waveform = sum of sine waves of all frequencies

$$x(t) = \int_{-\infty}^{+\infty} X(w) \exp(j\omega t) dw$$

デジタル信号処理 : ソフトウェア or ハードウェアによる信号データに対する処理

Digital Signal Processing : Processing of signal data by software or hardware

応用例 : TV (HDTV), CD, DVD, 電子楽器 (デジタルシンセ, ボーカロイド, 等), ..

Application Example : Tv(HDTV), CD, DVD, electronic musical instruments (digital synthesis, vocaloids)

デジタル信号の特徴 : デジタル化の際に多少の劣化はあるが, 伝送・記録に伴う劣化はほとんどない.

Characteristics of digital signals: Although there is deterioration during digitization, there is almost **no deterioration during transmission and recording**.

時間的にも強度的にも離散的 "discrete" → 情報量は有限 "finite"

Discrete in time and intensity → finite amount of information

情報量が減った (というか無限は取り扱いえない) 分だけ劣化...しかし, アナログ信号としてのパルス波形がある程度まで変化しても, デジタルデータとしては変化しない.

It deteriorates as the amount of information decreases (or rather, it cannot handle infinity).. However, even the pulse waveform as an analog signal changes to a certain extent, It does not change as digital data.

アナログ→デジタル変換(A/D変換)"analog-to-digital conversion": 符号化 "**encoding**"

Analog to digital conversion - encoding

PCM (Pulse Code Modulation)を利用. PCM is used

デジタル→アナログ変換(D/A変換)"digital-to-analog conversion": 復号化 "**decoding**"

Digital to analog conversion - decoding

PCM の逆変換. Inverse conversion of PCM

パルス符号変調 "pulse code modulation": アナログ信号波 →【標本化】→【量子化】→ 符号列 (ビット列)

Pulse code modulation: Analogue signal wave → [sampling] → [quantization] → code string (bit string)

A method used to digitally represent samples of analog data. Involves 3 steps, sampling, quantization, and coding

標本化 "sampling"

Sampling

信号の時間方向の離散化 "discretization": アナログ信号 $x(t)$ の一定時間毎の強度 (実数値) $X_k = (kt_s)$ を取得.

Discretization" of the signal in the direction of time: Get the intensity (real value) $X_k = (kt_s)$ of the analog signal $x(t)$ at regular time intervals

標本化周期 “sampling period” (or 標本化間隔 “sampling interval”) : t_s (or Δt)

Sampling period “sampling interval”: t_s (or Δt)

標本化周波数 "sampling frequency" (or "sampling rate") : $f_s = 1/t_s$ (1.1)

→ 標準的な標本化周波数は 44.1 kHz や 48 kHz.

The standard sampling frequency is 44.1 kHz or 48 kHz.

標本化周波数 f_s が小さすぎると、原信号 "original signal" に含まれていた情報の多くが失われてしまう。

If the sampling frequency f_s is **too small**, much of the information contained in the "original signal" is **lost**.

一方, f_s が大きいと、データ量も肥大化してしまう

On the other hand, when f_s is **large**, the amount of data is also **bloated**./excessive in amount.

→ 原信号を完全に再現可能となるような必要最小限の f_s の値は？

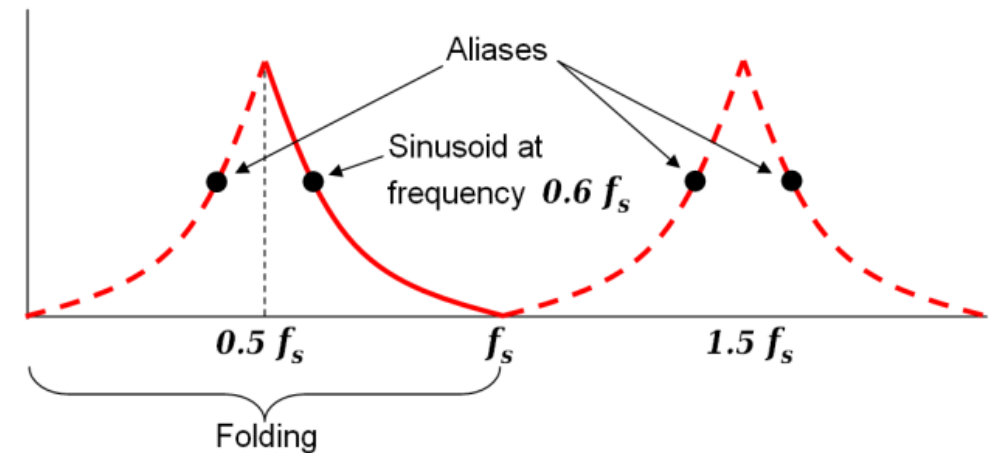
What is the minimum value of f_s necessary to allow full reproduction of the original signal?

ナイキスト-シャノンの標本化定理 "**Nyquist-Shannon sampling theorem**" (or "**sampling theorem**") : $f_s > 2f_{max}$ → 標本値から原信号を完全に再現可能.

Nyquist-Shannon sampling theorem" (or "sampling theorem"): → The original signal can be completely reproduced from sampled values.

$f_s / 2$: ナイキスト周波数 "**Nyquist frequency**"

Sometimes referred as **folding frequency**



https://en.wikipedia.org/wiki/Nyquist_frequency

f_{max} : 原信号のスペクトルの最高周波数.

f_{max} : The highest frequency of the spectrum of the original signal.

逆に、標本化周波数の半分より高い周波数が含まれていると、再現不可能. 偽信号 "alias" が混入してしまう

Conversely, if frequencies higher than half the sampling frequency are included, they cannot be reproduced. **False signal "alias" will be mixed in.**

通常は、標本化の前に、信号の高周波成分をカットしておく. え？カットしちゃっていいのかい？ はい、人間の可聴周波数範囲は 20 Hz～20 kHz なので無問題

Normally, the **high-frequency components of the signal** are **cut off** before **sampling**. What? Is it OK to cut them off? **Yes**, since the **human audible frequency range** is **20 Hz to 20 kHz**.

量子化 "quantization":

信号の強度方向の離散化: 強度 (実数値) $x \rightarrow$ バイナリデータ (整数値) x'

Discretization of signal intensity direction: intensity (real value) $x \rightarrow$ à binary data (integer value) x'

線形 PCM では $x' = \alpha x + \beta$

In linear PCM



量子化深度(量子化ビット数)"quantization bit depth"

Quantization depth (number of quantization bits)

標準的な深度は 16 bit → 65,536 階調. 深度は階調数 "number of gradation levels" そのものではありませんよ. 階調数の二進数の桁数-1です.

Typical depth is 16 bit → 65,536 gradations. Depth is not the "number of gradation levels" itself. It is the number of binary digits of the number of gradation levels -1.



量子化誤差 "quantization error"

深度が小さいと誤差は大きい(精度は低い). 一方, 深度が大きいと誤差は小さい(精度は高い)が, データ量が肥大化

Smaller depths have larger errors (lower accuracy). On the other hand, a large depth has a small error (high accuracy), but the amount of data becomes large

a. 線形PCM "linear PCM": 信号の瞬時値を均等間隔で量子化 標準的な手法

Linear PCM "linear PCM": quantization of the instantaneous values of a signal at **equal intervals**.

Standard method.

b. 非線形PCM "nonlinear PCM" 対数PCM "log-PCM": 信号の瞬時値を対数スケールで量子化. 感覚的な誤差を抑制しつつ、データ量の削減に効果的.

Non-linear PCM "nonlinear PCM", logarithmic PCM "log-PCM": Quantization of the instantaneous value of a signal on a logarithmic scale.

Effective for reducing the amount of data while suppressing sensory errors.

ウェーバ-フェヒナーの法則 "**Weber-Fechner law**" (or "**Fecher's law**") : 感覚量(心理的知覚)は物理量(物理的刺激)の対数に比例

Weber-Fechner law" (or "Fecher's law"): **sensory quantity** (psychological perception) is **proportional** to the **logarithm of the physical quantity** (physical stimulus).

人間の知覚のダイナミックレンジ "dynamic range" (識別可能な最小値～最大値の範囲)の広さの理由. 音量の単位「デシベル(dB)」もここから派生

→ Reason for the wide dynamic range of human perception "dynamic range" (the range between the minimum and maximum identifiable values). The unit of sound level, the decibel (dB), is also derived from this.

c.差分PCM, 適応差分PCM: 瞬時値自体ではなく, その変化量を量子化. Differential PCM, Adaptive Differential PCM: Quantizes not the instantaneous value itself, but the amount of change in it.

量子化誤差 and/or データ量の削減に効果的. Effective in reducing quantization errors and/or data volume.

PCM について

(1) アナログ信号をデジタル化しておくが、後で原信号を高精度に復元したい。原信号のスペクトルの最高周波数が 24 [kHz] の場合、デジタル化の際に設定すべき標本化周波数の最低値 [kHz]を求めよ。



Thank you for listening!